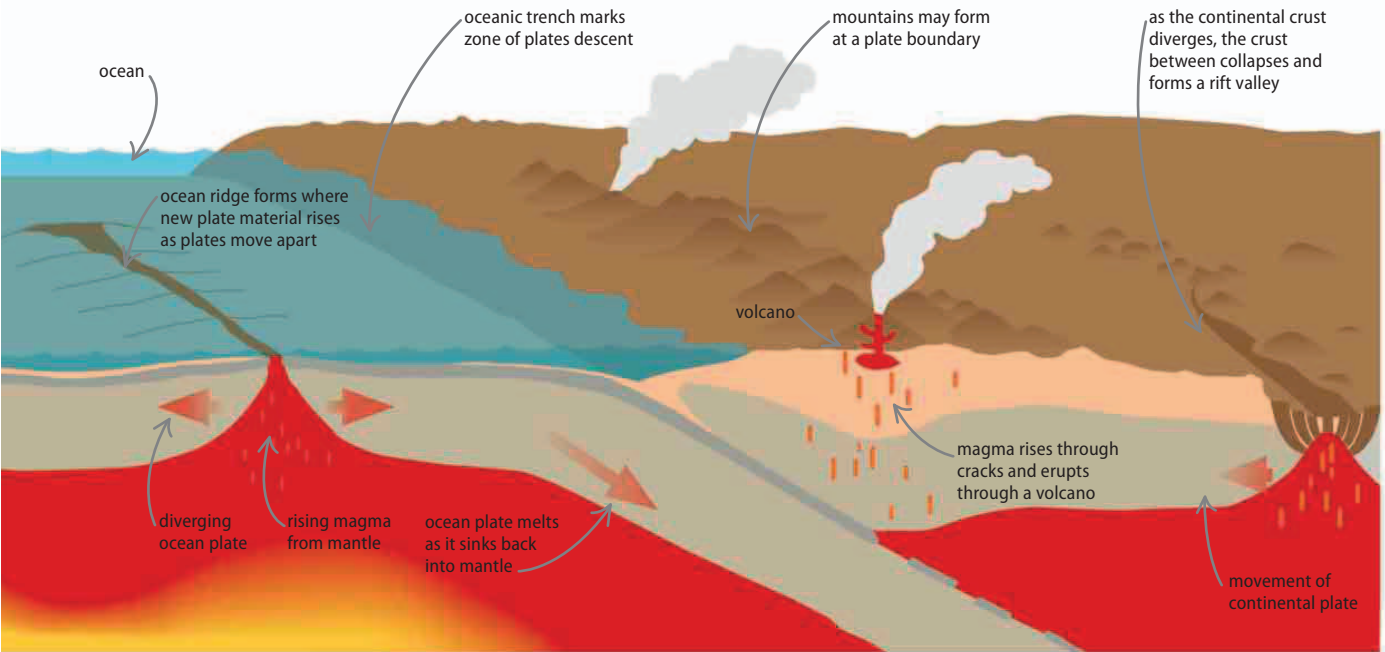


Plate tectonics

Earth's crust is broken up into sections called tectonic plates. The plates drift on the mantle as it is slowly churned by currents caused by heat from the core. Where two plates move together, at a convergent boundary, one plate dives under another to form a mountain range. At a divergent boundary the plates move apart and molten material from the mantle, known as magma, erupts at the surface as a volcano. Where plates grind alongside each other, earthquakes occur as the rocks catch and then jerk free.

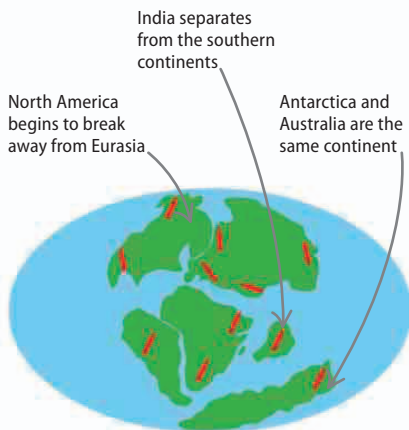
▽ Violent Earth

As plates constantly move, oceans are pulled apart, and continents may either crash into each other or break away.



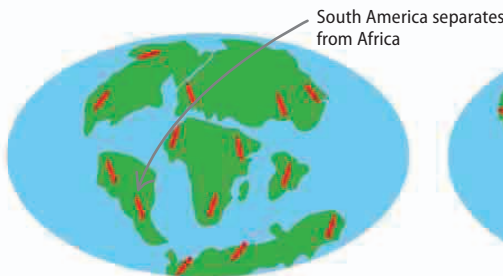
Continental drift

Over millions of years, the motion of Earth's plates has caused the continents to drift apart. If you could put them together, they would all fit, like a jigsaw puzzle. This idea is supported by matching patterns of rocks and fossils on lands now separated, and may explain why similar animals are found on opposite sides of the world.



△ 130 mya

At this point, North America begins to break up from Eurasia (the landmass comprising Europe and Asia). Australia and Antarctica are joined together.



△ 70 mya

Divergent plates continue to open up the Atlantic Ocean. South America drifts west, Antarctica heads for the South Pole, and India creeps towards Asia.



△ Present day

India is in place after colliding with the Eurasian mainland. Greenland separates from North America.